Gonzalo Martin Perez

Software Engineer | AI Engineer | Product Engineer

Generative AI · AI Agents · LLM · RAG · GraphRAG · Python · FastAPI · LangChain · LangGraph · Neo4j · Java · Spring Boot · TypeScript · Node.js · React · Next.js · AWS

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PROFESSIONAL SUMMARY

AI Software Engineer with an Information Systems Engineering degree. Build user-focused enterprise, fintech, and blockchain products end to end: AI agents and RAG/GraphRAG, distributed backends, web and mobile applications, and production operations. Reduced measured average mobile startup from about **7–8 s to 1–2 s** and contributed to ingesting **20M+ promotions**. Combine product judgment, maintainable architecture, performance, and security to turn complex requirements into reliable software.

PROFESSIONAL EXPERIENCE

AI Engineer | Rampy

Sep 2026 – Present | United States · Remote

AI Engineer · end-to-end product engineering · Work with three founders to bring AI, fintech, web, and mobile features from technical decisions through delivery and production operations.

- **Modernized the production application end to end**: refactored the backend into hexagonal architecture and organized the React/Next.js frontend by domain and feature; built Rampy's complete design system with reusable components. Implemented end-to-end encryption (E2EE) for user conversations, storing messages encrypted in the database.
- **Built and optimized production AI agents with Neo4j GraphRAG** using Agno, LangChain/LangGraph, and OpenAI API; integrated PostgreSQL/pgvector, Mem0, Vertex AI, and DeepInfra for retrieval, memory, and model access. Improved tool selection and evaluation: optimized flows ran at least 2× faster by my estimate, while removing redundant calls reduced token usage in affected queries by an estimated 30%.
- **Cut measured average mobile app startup from about 7–8 s to 1–2 s**; shipped React Native, Kotlin and Swift features, using Android Studio for Android. Implemented end-to-end event tracking with Firebase Analytics, Clarity Mobile on every screen, Singular SDK and Meta App Events, avoiding duplicates.
- **Integrated Morpho, Aave, and Compound end to end** for positions, yields, deposits, and withdrawals across backend, frontend, and transaction execution, with completeness checks and webhook-based status tracking; also worked with Hyperliquid and LI.FI, Privy authentication and wallets, and Stripe integrations, all in production.
- **Built a consolidated portfolio and 24-hour returns view** with parallel calls, prefetching and Redis caching; comparable vault, market and perpetual queries moved from seconds to milliseconds.
- **Cut validation, build and deployment from about 1 hour to 10 minutes**, retaining checks and preparing DigitalOcean environments. Built the enterprise backoffice for application management, monitoring, telemetry and status; continue to expand its capabilities.

Software Engineer | Teamcubation

Dec 2025 – Aug 2026 | Argentina · Remote

Built software with PostgreSQL and MongoDB for Payway's merchant promotions platform, in a 13-country ecosystem with thousands of new promotions daily.

- **Built the merchant-facing Promotion Assistance System** with LangChain/LangGraph, Neo4j, PostgreSQL/pgvector and OpenAI API. GraphRAG related enterprise policies and promotions; its internal harness orchestrated agents, evaluated answers, tuned performance and enforced domain and data-leakage guardrails. Connected to the Spring WebFlux BFF for portal access, the assistant answered queries without executing commercial actions.

- **Built event-driven ingestion for 20M+ historical promotions in production**: S3 → SQS → Python/FastAPI Lambda → Java/Spring Boot and Node.js/NestJS microservices, each writing to its own PostgreSQL database. Built Python/Node.js tooling to reproduce local loads, reconcile results and identify failed promotions.
- **Delivered the promotions experience end to end**: connected a Single-SPA React/TypeScript microfrontend, organized by feature and using a design system, to Java/Spring Boot and Node.js/NestJS microservices through a Spring WebFlux BFF; Amazon API Gateway bridged the HTTPS portal to internal HTTP services.
- **Built Java/Spring Boot and TypeScript/Node.js/NestJS microservices** using layered architecture, and developed and diagnosed containerized workloads running in Docker/Kubernetes-managed environments.
- **Created a custom agentic engineering workspace** for the product's multi-repository integration landscape to streamline context retrieval, token usage, and consistent feature delivery.

AI Engineer & Full-Stack Developer | Self Employed

`Jan 2024 – Present | Argentina · Remote`

Independent Product Engineering · Revenue-generating products, internal operations systems, and AI automation for e-commerce, social media, and digital-product clients.

- **Delivered Pequeverso's digital-product storefront** with Next.js on Hostinger and an informational product and shopping assistant. Built stores, APIs and Python/Node.js automation under Spec-Driven Development, cutting tasks from hours to minutes.
- **Built MCP-, Claude-, and LLM-powered pipelines and agents** for end-to-end content research, generation, and publishing; clients estimated operating-cost reductions of at least **70%**.
- **Designed and shipped digital products, backoffices, and inventory/point-of-sale extensions** that generated **USD 10,000+ in cumulative client profit across multiple projects**, translating business needs into maintainable software.

Software Engineer | Cooperativa Obrera

`Dec 2024 – Dec 2025 | Bahía Blanca, Argentina · On-site`

Permissions Management System · Took a primary role in designing and building an enterprise backoffice for authorized personnel at an organization with **150+ branches**, **2.7M+ members**, and 50+ internal systems.

- **Designed and built a permissions back office integrating 10+ enterprise systems** with different databases and permission models; automated roughly 10 minutes of manual permission lookup into a workflow taking under 10 seconds, giving administrators a consistent experience.
- **Standardized system onboarding** with reusable structures, filters, and validation, reducing permission-related tickets and manual assignments by at least **80%** and eliminating at least **95%** of recurring assignment bugs, based on before/after tickets and logs.
- **Built a responsive, dynamic Next.js/TypeScript frontend**, organized by feature and using a design system, connected exclusively to a Python/FastAPI BFF. The backend used layered architecture; the BFF standardized frontend views and connected Java/Spring Boot, Node.js/NestJS and PHP services through OpenAPI/Swagger; microservices used MySQL / MariaDB databases.
- **Kept critical endpoints below 300 ms on average** in repeatable tests on enterprise on-premise servers through SQL tuning, indexing, and pagination. Built an agentic engineering workspace to preserve context across integration repositories.
- **Implemented LDAP, RBAC, and an exportable audit trail** to control access and trace permission changes; automated bulk data loading and validation with Python in a MySQL/MariaDB and Redis environment.

TECHNICAL SKILLS

Applied AI & Agentic Systems: AI Agents, RAG, GraphRAG, LangChain, LangGraph, Context Engineering, Prompt Engineering, Embeddings, Chunking, pgvector, Neo4j, LLM Evaluation, AI Testing, AI Guardrails, Fine Tuning, LLMOps, Model Context Protocol (MCP), OpenAI API, Agno, Claude API

Backend & APIs: Spring Boot, FastAPI, NestJS, Spring WebFlux, Spring Security, Laravel, REST APIs, Microservices, Celery, OpenAPI

Languages & Runtimes: Python, TypeScript, Java, JavaScript, Node.js, SQL, Kotlin, Swift, PHP, Solidity

Web & Mobile: React, Next.js, React Native, Android Studio, Design Systems, Feature-Based Architecture

Data & Architecture: PostgreSQL, MySQL, MariaDB, MongoDB, Redis, Hexagonal Architecture, Layered Architecture, Event-Driven Architecture, SQL Tuning

Cloud & Delivery: DigitalOcean, AWS Lambda, Amazon API Gateway, Amazon S3, Amazon SQS, CloudWatch, Docker, Kubernetes, GitLab CI/CD, GitHub Actions

Quality, Delivery & Observability: JUnit, Mockito, SonarQube, OpenTelemetry, Prometheus, Grafana, RBAC, LDAP, Application Security, End-to-End Encryption, Infisical

Fintech & DeFi Integrations: Fintech, Blockchain, Web3, DeFi, Morpho, Aave, Compound, Hyperliquid, LI.FI, Privy, Stripe, Swaps, Vaults, Perpetuals, Non-Custodial Wallets

EDUCATION

Information Systems Engineer | Universidad Nacional del Sur (UNS)

`Jan 2020 – Oct 2025 | Bahía Blanca, Argentina | Average grade: 8.67 / 10`

Graduated from a 5-year accredited engineering program. Completed 34/34 courses, an **8.67/10 average**, a **10/10 Final Year Project**, and Supervised Professional Practice. UNS ranked **#10 nationally** in SCImago Institutions Rankings 2026. Relevant coursework: Artificial Intelligence, Systems Architecture & Design, Databases, Software Verification & Validation, Algorithms & Complexity, and Formal Methods.

View education and academic record →

SELECTED PROJECT

Filomena · Production Exam Platform for Health Sciences

`Jan 2025 – Dec 2025`

Principal author and contributor in a three-person team building a production platform active at **5 Argentine national institutions**; Final Year Project graded **10/10**.

- **Supported 1,000+ concurrent users observed in production** while preserving data consistency and keeping critical endpoints below **300 ms on average**, using SQL optimization, Redis caching, and queues.
- **Led architecture, backend, frontend, and infrastructure**, modernizing a CakePHP + jQuery monolith into an API-first product with a Laravel REST API and Next.js/React/TypeScript frontend.
- **Reduced institutional onboarding from hours to minutes** with production Docker/GitHub Actions deployments observed below **15 minutes**.
- **Implemented production security and observability** with Prometheus/Grafana, multi-role RBAC, student anonymity and per-institution data isolation.
- **Coordinated Agile delivery** from requirements through production.

View case study →

LANGUAGES

Spanish: Native

English: Professional working proficiency (B2)

CERTIFICATIONS

Docker Fundamentals - Universidad Nacional del Sur | Mar 2024

Professional Power Skills - Kognité | Jun 2026